Tiago Ferrão-Lindqvist

Data Engineering | Test Data, Python & BI

PROFESSIONAL SUMMARY

Test & Validation Engineer with experience building and supporting Python-based test-data processing and reporting tools, SQL Server staging workflows, and Power BI dashboards. Processes approximately 2 GB of rig data per week across six rigs; the processing tool is used by four test engineers, and the dashboard by about 15 engineers in validation and design. Brings a mechanical testing and root-cause analysis background, with ongoing data engineering study.

TECHNICAL SKILLS

Programming and data processing: Python, pandas, NumPy, SQL, MATLAB, Simulink
Databases and reporting: SQL Server, SQLite, Power BI
Data acquisition and visualisation: TDMS, CSV, npTDMS, Matplotlib
Tools: Windows Task Scheduler, Flask, Docker Compose, Git
Testing: Durability and vibration testing, test-data processing, test reporting, root-cause analysis

PROFESSIONAL EXPERIENCE

Test & Validation Engineer | Corvane Automotive Components | 2019–Present

- Conduct durability and vibration testing and contribute to root-cause analysis.
- Develop test-data tooling and reporting workflows for rig data.
- Trained three apprentices in rig safety and test setup (2022–2024).

Mechanical Design Engineer | Halvik Pumps | 2014–2019

- Designed pump casings and impellers and ran structural FEA.
- Supported performance test-bench activities.

SELECTED PROJECTS

Rig Data Processing and Reporting Tool | Corvane Automotive Components | 2021–Present

- Developed Python scripts to parse TDMS and CSV output from six rigs, processing about 2 GB of data per week.
- Flagged channels outside test specifications and generated report figures using pandas, NumPy, Matplotlib, and npTDMS.
- Automated nightly runs with Windows Task Scheduler on a lab PC and wrote summary results to an IT-created staging table in SQL Server.
- Packaged the tool for internal use; four other test engineers use it for their tests.

Test Results Dashboard | Corvane Automotive Components | 2023

- Built a Power BI dashboard on the staging table to show pass/fail trends and cycles to failure by product generation.
- Used by about 15 engineers in validation and design.

Adjuster Drive Load-Profile Model | Corvane Automotive Components | 2020–Present

- Developed a MATLAB/Simulink model of the seat-adjuster drive, including its motor, gearbox, and rail friction, to select rig load profiles matching customer duty cycles.

Bike-Sharing Trips API | Personal Learning Project | 2025

- Built a small Flask API with three read-only endpoints over a public bike-sharing dataset loaded into SQLite; it runs locally on a laptop.
- Followed a tutorial to run the project with Docker Compose.

PROFESSIONAL DEVELOPMENT

Online Data Engineering Course | In progress; six of nine modules completed. No certificate.